DAVID SÁNCHEZ-FUENTES

Data Mining Psychological Diagnoses in Chihuahua

Identifying vulnerable help-seeking populations through demographic comparison, statistical analysis, and mixed-type clustering.

Analysis of 18,220 Municipal Consultation Records · Chihuahua City, Mexico



CHIHUAHUA · CHIH



CONTEXT

Aggregate prosperity can mask concentrated distress

13.7

Suicides per 100,000 in Chihuahua state, the highest in Mexico versus **6.4 nationally**

The Core Question

Which demographic cohorts and diagnostic groups appear most prominently in front-line public psychological services?

Primary Dataset

18,220 consultation records maintained by the Chihuahua municipal government, capturing real-world clinical intake data across diverse centers.

INEGI & SECRETARÍA DE SALUD COMPARATIVE BASELINE

RELATED WORK

Related work

What prior studies measure—and what this project adds

Oropeza-Acevedo (2025)

A Chihuahua hospital study of self-harm admissions examining sobriety, physical illnesses, and psychiatric disorders—an acute hospital setting.

Fernández-López et al. (2021)

Each 1ºC rise in temperature was associated with a **2.8% increase** in suicide rate from 2008–2018—a mortality pattern.

This project

It studies psychological consultation records rather than hospital self-harm admissions. Monthly temperature showed no statistically significant association with consultation volume (**p = 0.56**), suggesting completed-suicide seasonality does not directly translate to help-seeking behavior.

RESEARCH FRAMEWORK

The project follows four analytical questions

01 **Who is seeking psychological help?**

02 **How does that profile differ from Chihuahua's broader population?**

03 **Which variables move together, and does temperature predict monthly volume?**

04 **Can mixed demographic and diagnostic profiles reveal interpretable vulnerable subgroups?**

METHODS · EVALUATION

Evaluation

A transparent validation framework for inference, comparison, and reproducibility.

Inference

Statistical significance set at **α = 0.05**. Pairwise post-hoc comparisons use a Bonferroni adjustment.

Seasonality check

Match **18 monthly observations** (Jan 2025–Jun 2026) with historical mean temperatures from Metar-Taf.com.

Cluster validation

Compare silhouette scores for **k = 3–10** and inspect a UMAP projection built from Gower distances.

Reproducibility & benchmarks

Fix random seed at **42**. Benchmark population structure against the Data México Chihuahua population pyramid and published census education proportions.

DATA PREPROCESSING

From messy administrative records to analyzable features

01

Raw Ingestion

18,220 consultation records with uncurated text entries, transcription typos, duplicates, and missing demographic fields.

02

Diagnostic Mapping

30+ inconsistent diagnosis strings normalized into **28 canonical clinical labels** aligning symptom definitions.

03

Socioeconomic Binning

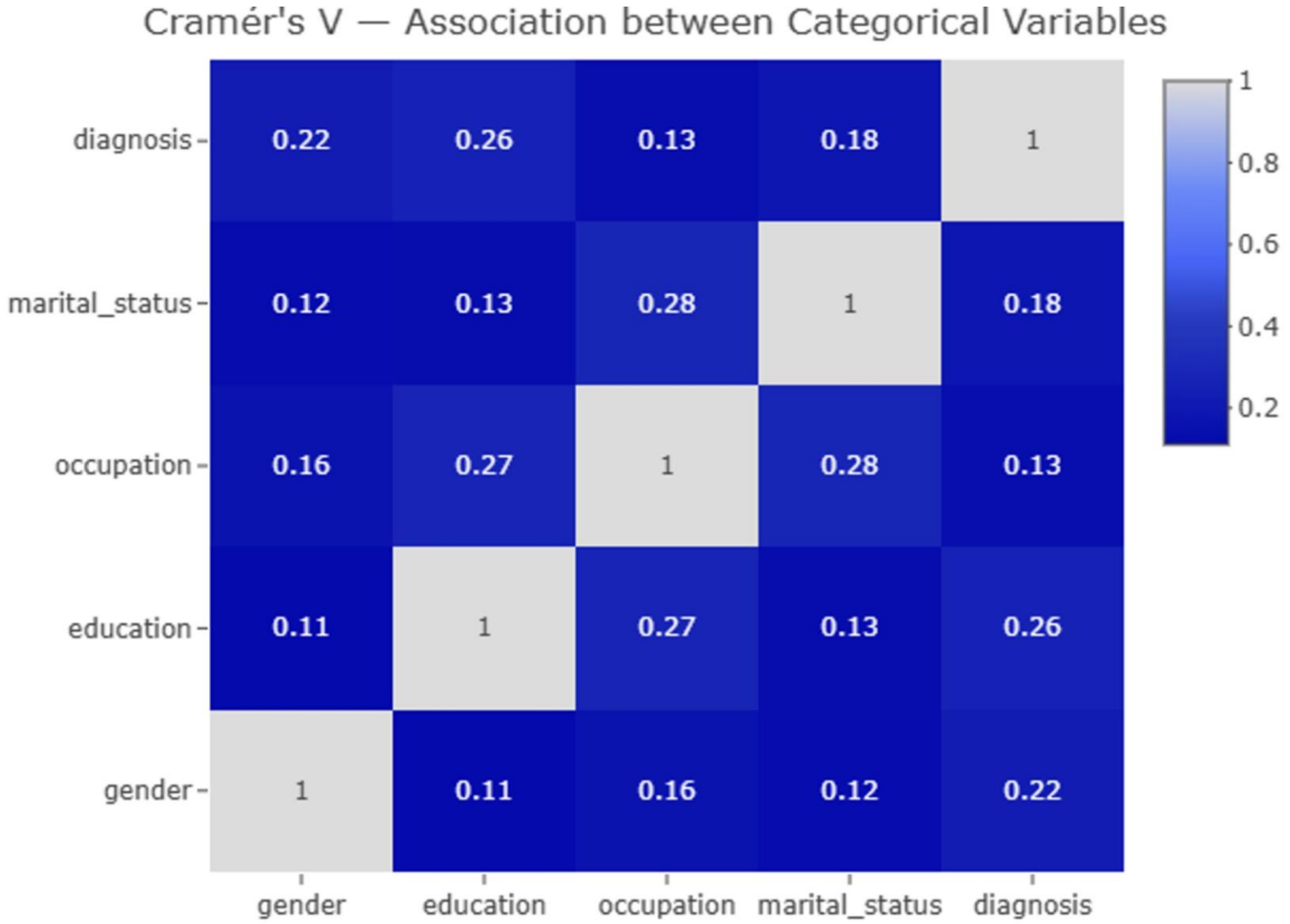
20+ occupation categories consolidated into 11 classes; education encoded into 5 ordered levels to stabilize sparse cells.

Feature harmonization pipeline: resolving administrative heterogeneity without truncating rare diagnostic signals.

ASSOCIATION ANALYSIS

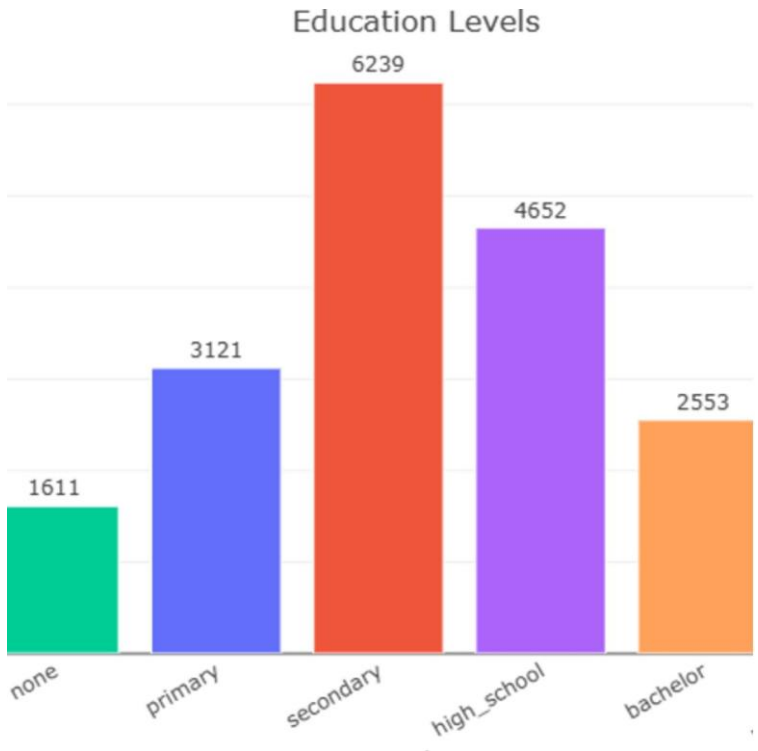
Categorical associations are modest.

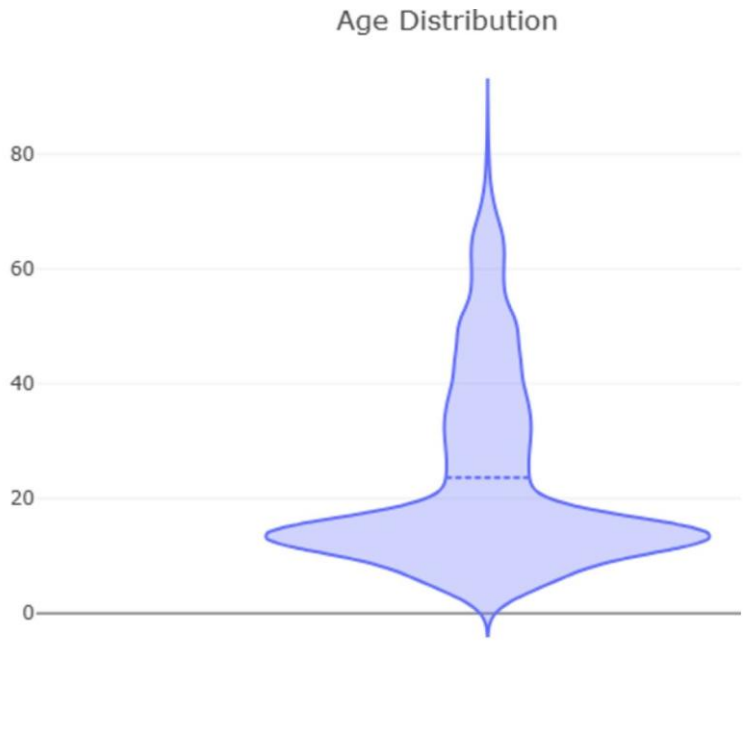
No single categorical variable dominates the dataset.

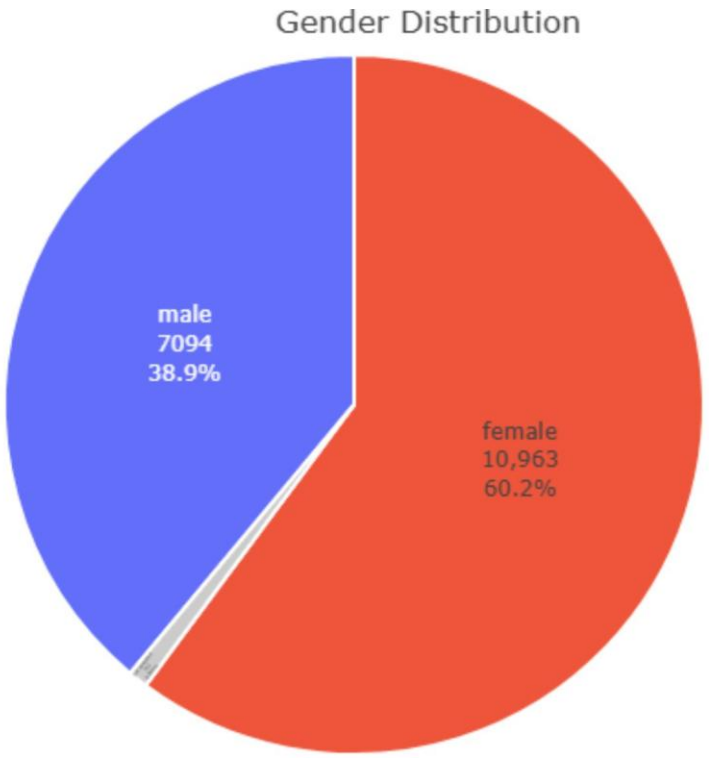


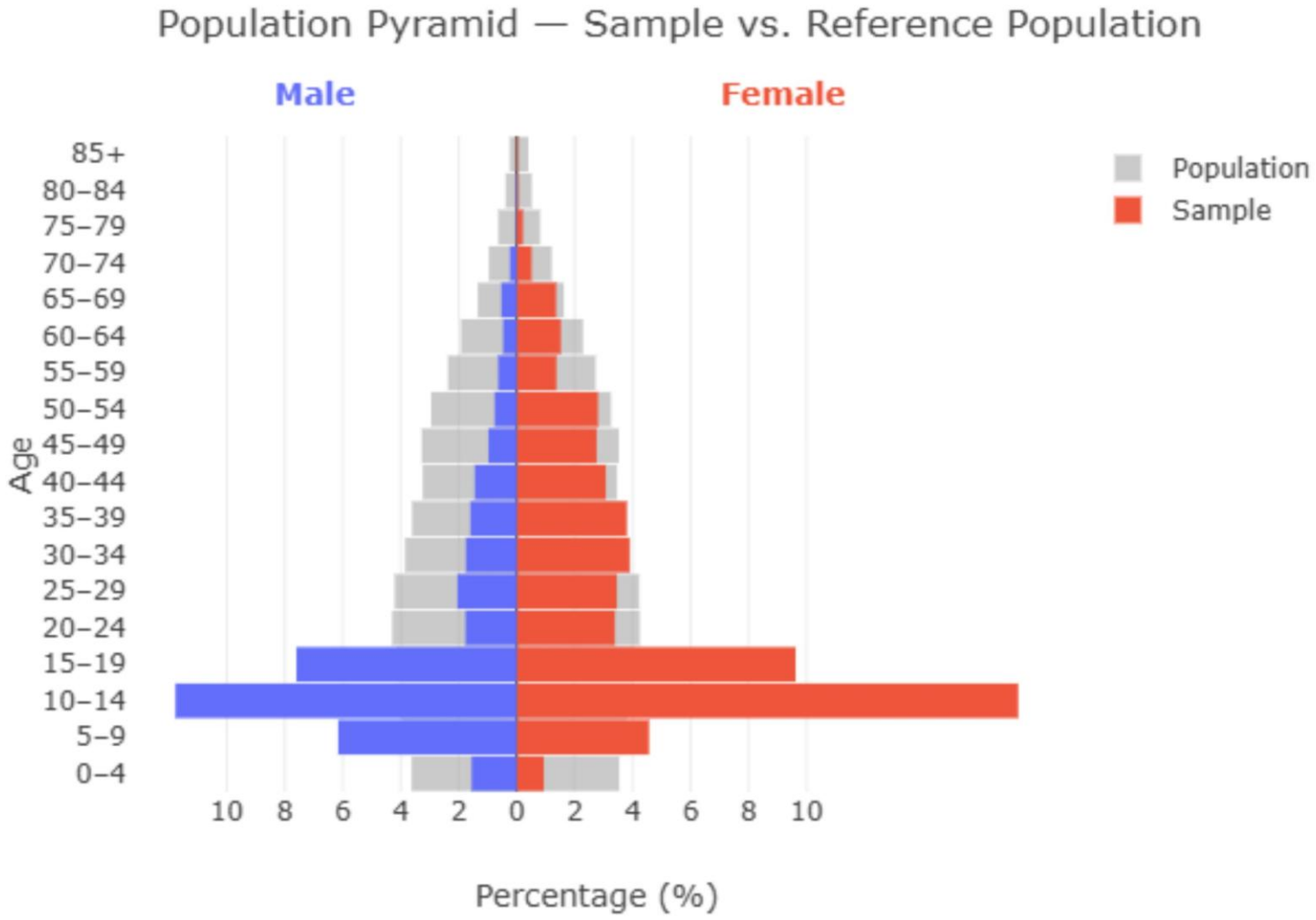
Sample profile

Who appears in the help-seeking sample





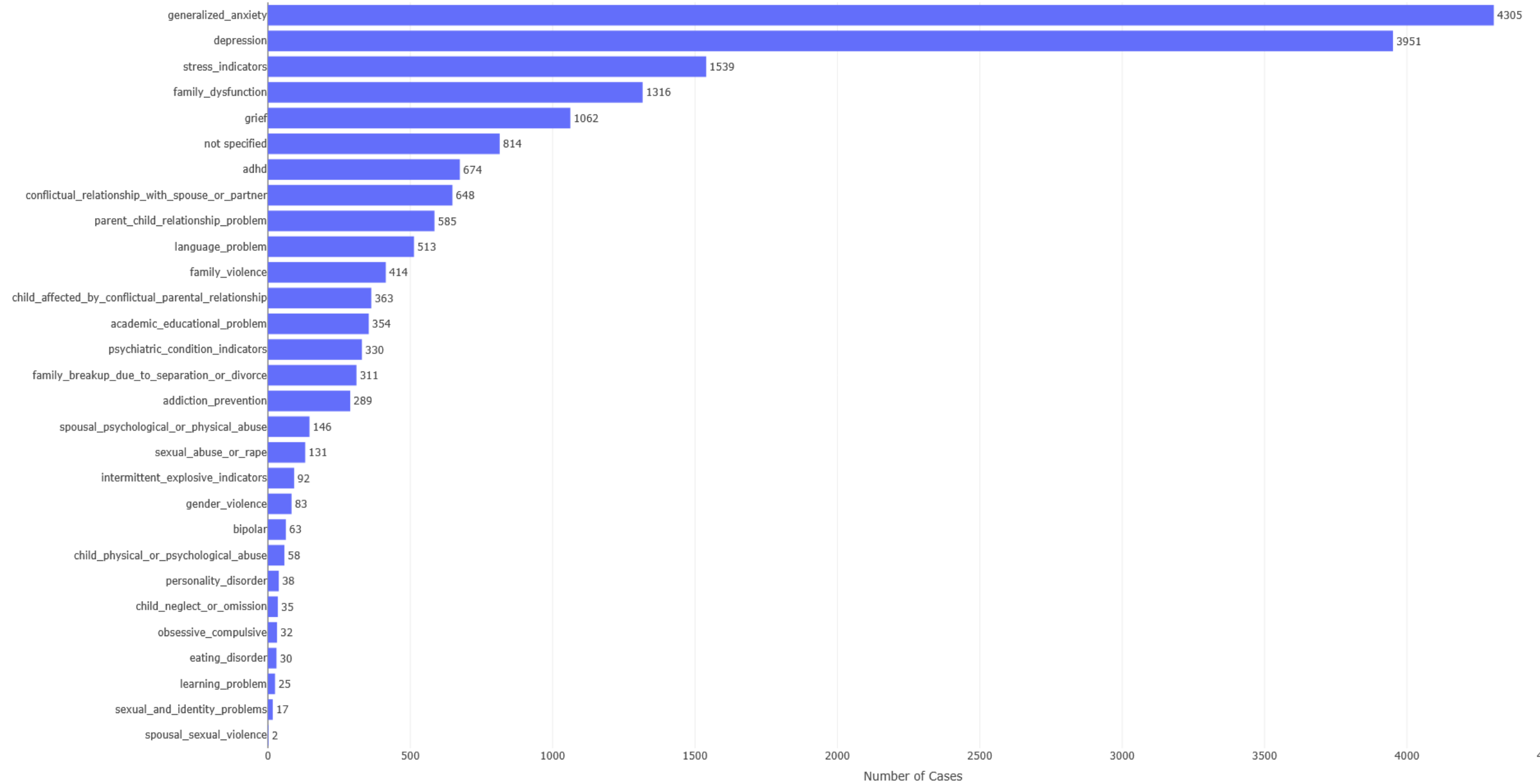




The sample diverges sharply from the reference population

Help-seekers are over-represented at ages 10–19, especially males aged 10–14, while adults aged 25–49 are under-represented.

Diagnosis counts



METHODOLOGICAL PIVOTS

The method changed when the data demanded it

CLUSTERING ARCHITECTURE

k-means → k-prototypes

Challenge: The clinical dataset combines continuous age with nominal diagnosis, sex, and occupation fields. One-hot Euclidean distance artificially inflates categorical dimensionality and distorts geometric centers.

k-prototypes combines Euclidean distance for numeric features with matching dissimilarity for categorical features via a calibrated cost penalty.

MANIFOLD EMBEDDING

PCA → UMAP on Gower

Challenge: Standard PCA assumes linear continuous covariance, forcing arbitrary binary projections that collapse distinct qualitative clinical syndromes into uninterpretable clouds.

Gower dissimilarity handles mixed types natively, allowing UMAP to preserve non-linear local neighborhood topology without numeric distortion.

Rigorous data science adapts analytical tools to feature geometry and sample limitations rather than forcing mismatched defaults.

UNSUPERVISED MIXED-TYPE CLUSTERING

Six clusters organize help-seeking by life stage and concern

CLUSTER 1

Middle-aged adults with grief

Mean age: 52

CLUSTER 2

Children with ADHD

Mean age: 9

CLUSTER 3

Male high-schoolers with generalized anxiety

Mean age: 17

CLUSTER 4

Secondary students with depression

Mean age: 13

CLUSTER 5

Female high-schoolers with depression

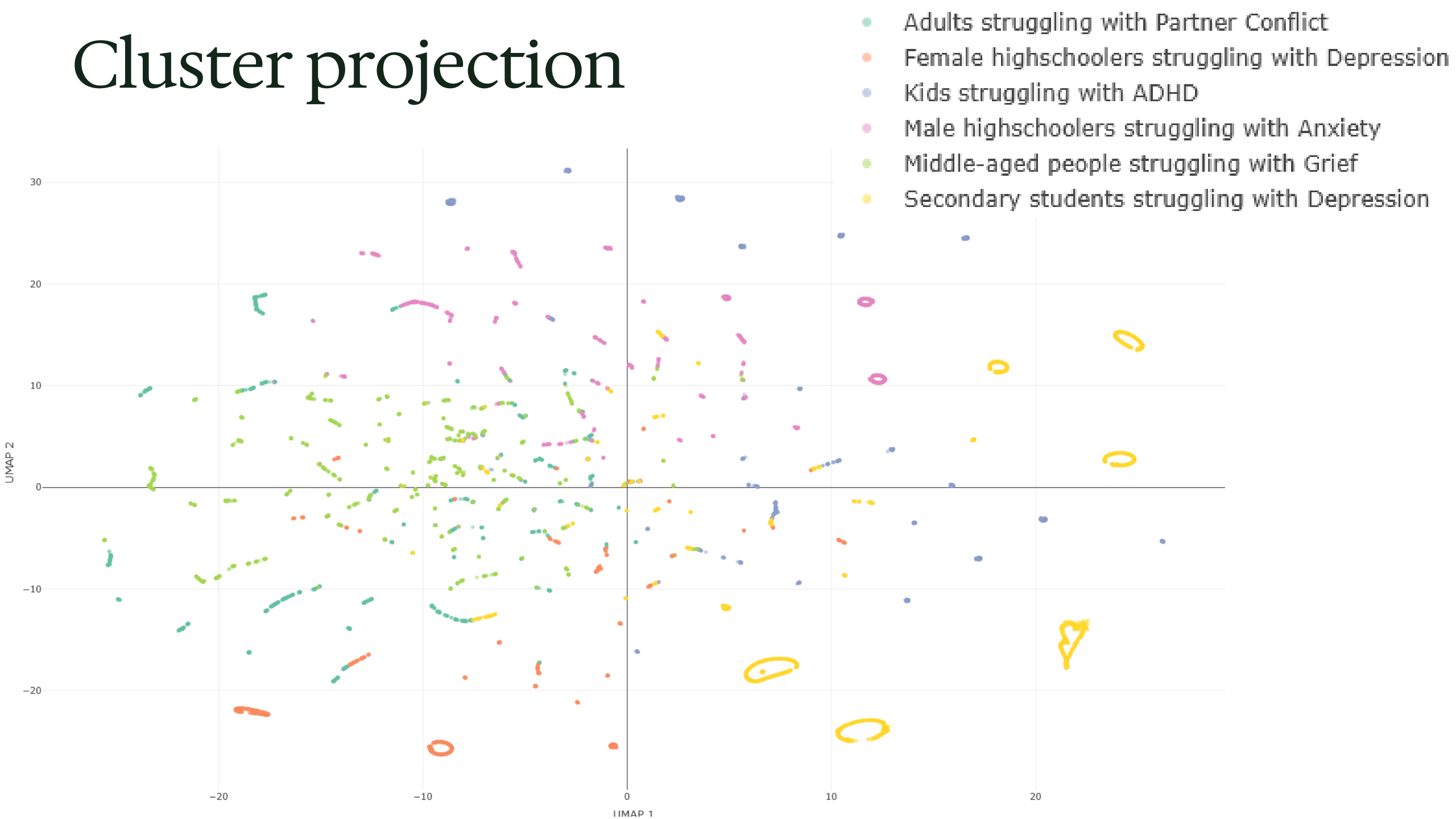
Mean age: 17

CLUSTER 6

Adults with partner conflict

Mean age: 37

Cluster projection



PROJECT PROGRESSION

Timeline

From proposal to presentation

01 PROPOSAL

Project proposed

02 DATA MINING

18,220 records mined

03 PREPROCESSING

Records cleaned and coded

04 EXPLORATION

Data explored

05 TESTING

Correlations and hypotheses tested

06 CLUSTERING

Help-seeking profiles grouped

07 VISUALIZATIONS

Findings made visible

08 FINAL OUTPUT

Report and presentation completed · Oct 2, 2026

METHOD · FRICTION POINTS

Challenges

Four decisions shaped the analysis

LABELS

Inconsistent diagnoses

30+ Spanish strings, accents, and typos required normalization.

EDUCATION

Rare categories

Kindergarten and post-doc levels were merged into adjacent categories.

AGE

Comparable bins

Age ranges were carefully binned to match all 18 reference-population groups.

CLUSTERS

Method switch

Poorly separated k-means clusters motivated the move to k-prototypes.

REFLECTION

Lessons learned

What the analysis made clear

01

Data type drives method choice

Mixed data called for a method that could respect both numeric and categorical structure.

02

Effect size beats p-value alone

At **n = 18,220**, Cramér's V helps prevent statistical significance from being over-interpreted.

03

Negative results are results

The non-significant temperature effect is informative, not a dead end.

So

SELECTED SOURCES

References

Works and datasets used in the project

Fernández-López et al. “Tendency and seasonality of suicide in Chihuahua, México.” *Salud Mental*, 2021.

Oropeza Acevedo. “Factores de riesgo asociados al intento suicida…” Universidad Autónoma de Chihuahua, 2025.

Instituto Municipal de Prevención y Atención a la Salud. Atención Psicológica dataset.

Metar & Taf. Weather in Chihuahua.

Data México. Chihuahua population pyramid.